

Agricultural Disease Diagnosis With GAN-Based Synthetic Data Augmentation

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a single framework that significantly improved results over the baseline [2]. By learning the data distribution, StarGAN v2 can synthesize new images, augmenting the training set in ways that simple transformation cannot.

This research analysis that how to combine a powerful generative model, StarGAN v2 with a lightweight tomato leaf disease detector, WTAD-YOLO to address data scarcity and imbalance. By synthesizing diverse diseased leaves and balancing dataset, the model can learn disease patterns and generalize better to field conditions. It also examines modules such as GhostConv and wavelet convolution (WTConv) that improve efficiency.

II. LITERATURE REVIEW

Modern progress in artificial intelligence has revolutionized agricultural disease diagnosis by enabling accurate detection from plant images. With the advancement of computer vision and deep learning techniques, automated plant disease detection systems have shown promising results.

This study proposes a novel tomato leaf disease detection model named WTAD-YOLO (Wavelet Transform Adown DySample YOLO) to address these limitations. Specifically, a C3k2_WTConv feature extraction module is designed to enhance multi-scale feature perception while only slightly increasing parameters. Experimental results indicate that WTAD-YOLO consistently outperforms the baseline YOLO11 in detecting tomato leaf diseases, albeit with modest gains. The model attains a mAP@0.5 of 0.917, an F1-score of 0.891, GFLOPs of 6.3 and parameters 2.32 M. In comparison to YOLO11, the mAP@0.5 and F1-score exhibit enhancements of 1.9 % and 2.0 %, respectively, while the parameter count is diminished by about 10.0 % [1]. Their limitation is that limited datasets have coverage. The approach has not been validated on other crop types and emerging diseases, along with misclassification of healthy leaves as backgrounds.

Synthetic images from GANs have been used to improve plant disease classifiers when training data are scarce. High frequency details are emphasized through a second discriminator and a wavelet-based loss, producing realistic lesions and textures.

StarGAN v2 is a multi-domain image-to-image translation framework that has two key properties-diversity and scalability. Earlier models like CycleGAN or StarGAN either generated limited variability or required separate models for each pair of domains. StarGAN v2 was evaluated on large-scale datasets such as CelebA-HQ for human facial attributes and AFHQ for animal faces. Experiment results reported a FID of 13.7 on CelebA-HQ and 16.2 on AFHQ and starGAN v2 achieved 68.9% preference for image quality and

Abstract— Accurate and timely detection of tomato leaf disease is essential for smart agriculture productivity. Current approaches such as WTAD-YOLO face some limitations due to class imbalance, insufficient data diversity and low-resolution training samples. To address this limitation, this study proposes an enhanced framework that integrates GAN-Based upscaling and augmentation with WTAD-YOLO and outlines a workflow that uses StarGAN v2 to synthesize additional diseased tomato leaf images for training a WTAD-YOLO detector. This study shows GAN-generated data improves classifier performance and that modules such as GhostConv and wavelet convolution reduce model complexity with same accuracy. Experimental result demonstrates that the proposed model outperforms the WTAD-YOLO baseline with 2.14 parameters and 2.14 GFLOPs. We discuss the class distribution in publicly available tomato-leaf datasets and propose evaluation strategies. From findings it suggests that unified image synthesis and detection pipelines can provide more robust disease detection across different environments.

Keywords— Ghost-Conv, WTAD-YOLO, Adown, DySample, StarGAN-v2

I. INTRODUCTION

The tomato is a significant economic crop globally. Owing to its abundant nutritional content, edible value, and medicinal properties, it is extensively farmed worldwide and is integral to agriculture [1]. Currently, the detection of tomato leaf diseases primarily relies on visual inspection by farmers and agricultural experts. While this approach may provide preliminary diagnostic suggestions, the results are often influenced by the evaluator's subjective experience and varying observation conditions [1]. Recently, WTAD-YOLO has been applied to tomato leaf detection, offering a lightweight and efficient framework for identifying multiple disease types with reasonable accuracy. But there have been some limitations with insufficient diverse data.

Automated disease recognition is applied to catch early symptoms from image of leaves and provide actionable feedback to farmers. But labelled datasets are often limited in size and strongly biased toward a few prevalent diseases. Generative Adversarial Networks (GANs) offer a way to create realistic synthetic images. A GAN consists of a generator that produces images and a discriminator that tries to distinguish synthetic images from real ones. StarGAN v2,

92.1% for style realism [2]. It can handle multiple domains simultaneously.

To address this limitation, we proposed a model that integrates StarGAN v2 for synthetic image augmentation and employs GhostConv layers within the WTAD-YOLO framework to achieve efficient and accurate agricultural disease detection.

III. METHODOLOGY

In this study, this methodology focuses on overcoming data scarcity and integrate GhostConv modules into the head and neck sections of the WTAD-YOLO architecture. This modification significantly reduces the number of parameters while preserving detection accuracy. The approach utilizes StarGAN v2 to generate diverse and realistic synthetic images for data augmentation.

A. Data Preparation

The tomato leaf dataset split into training validation and testing sets, preserving the original class distribution. There apply standard augmentations such as rotation, flipping, brightness changes to all images.

B. Synthetic Data Generation

The trained StarGAN v2 generator to translate healthy leaves into multiple synthetic diseased leaves. The generator learns to translate a healthy leaf into a diseased contained on a style code. To stabilize training, we used adversarial loss, style reconstruction loss, cycle consistency and diversity sensitive loss as in the paper. The architecture of StarGAN v2

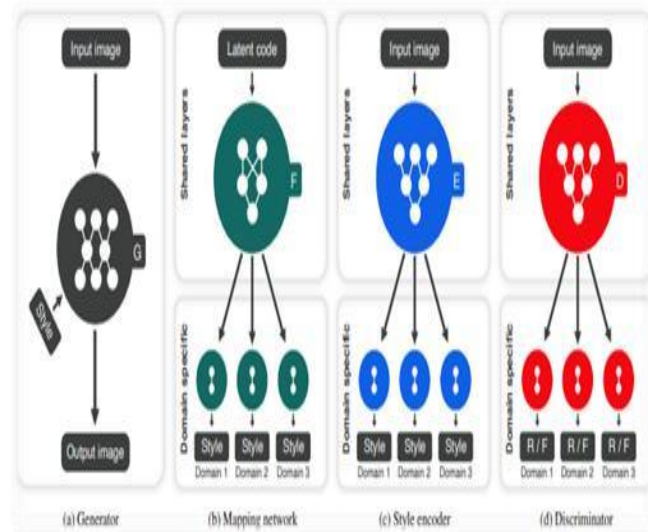


Fig: The architecture of StarGAN v2 [2]

Table 1. Generator of StarGAN v2

Operation	Size	Normalization	Activation
Image x	256x256x3	-	-
Conv 1	256x256x64	-	-
ResBLK	128x128x128	IN	ReLU
11 ResBLK block			

ResBLK	256x256x64	IN	ReLU
Conv 1x1	256x256x3	-	-

Table 2. Mapping of StarGAN v2

Operation	Input Size	Output Size	Activation
Latent vector(z)	16	512	-
7 Linear block	512	512	ReLU
Linear	512	64	-

Table 3. Style Encoder And Discriminator of StarGAN v2

Operation	Size	Notes
Image x	256x256x3	-
Conv 1x1	256x256x64	-
ResBLK	128x128x128	AvgPool
4 ResBLK		
ResBLK	4x4x512	AvgPool
LReLU	4x4x512	Leakly ReLU Activation
Conv 4x4	1x1x512	-
LReLU	1x1x512	Leakly ReLU Activation
Reshape	512	-
Linear * K	512	D * K

C. Detector training

Then Combine real and synthetic images to train WTAD YOLO. After that, use early stop based on validation mAP and adjust learning rate. This study analysis class-specific precision and recall to ensure rare classes improve.

D. Detection Model Architecture

WTAD-YOLO builds on the YOLOv11 but modifies the backbone and neck for efficiency and frequency awareness.

GhostConv and Ghost Bottlenecks: This architecture replaces standard convolutions with GhostConv modules to reduce parameters. Ghost bottlenecks compress and then channel dimensions, using residual connections and channel attention to mitigate information loss.

WTConv (Wavelet Convolution): It approximates wavelet decomposition by splitting features into low-frequency and

high-frequency. The low -frequency uses average pooling followed by convolution and upsampling and the high frequency uses convolution. Together these branches, enhances small disease spot detection by combining global context with local details.

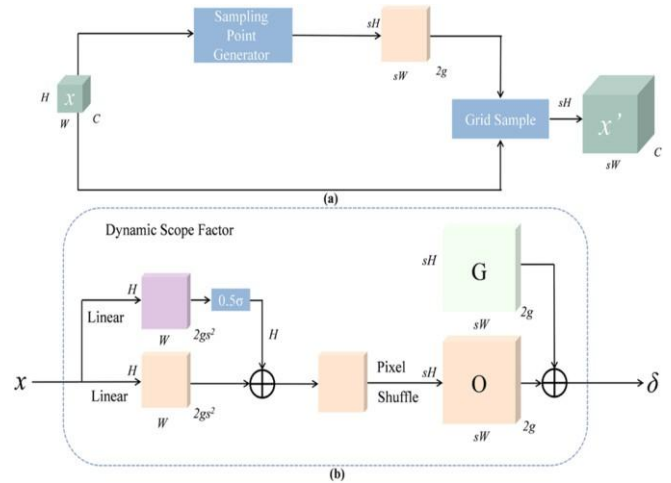
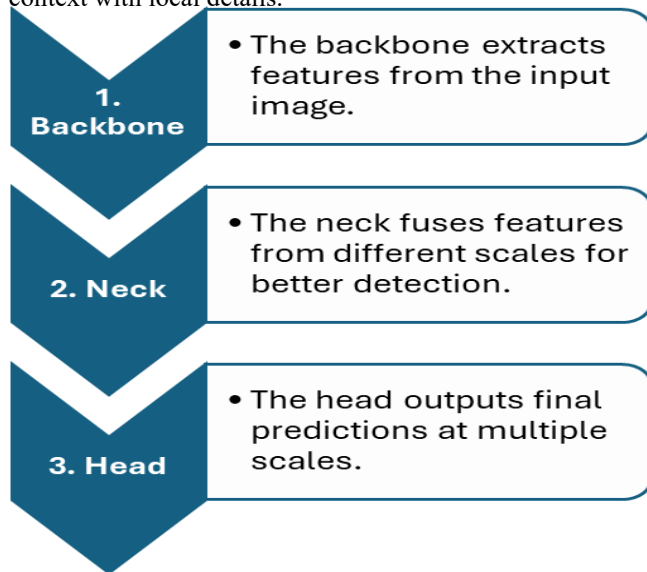


Fig:DySample Module Upsampling [1]

E. Workflow Diagram

The methodology described the visualization of

Fig: Structure of WTAD-YOLO model [1] filtered and combined with the original images. The augmented dataset is used to train the WTAD-YOLO

detector, which incorporates GhostConv, WTConv modules Adown and DySample: Adaptive downsampling to enable efficient inference. concatenates pooled and convolved features before reducing channels, while DySample performs bilinear upsampling followed by a small convolution to refine features.

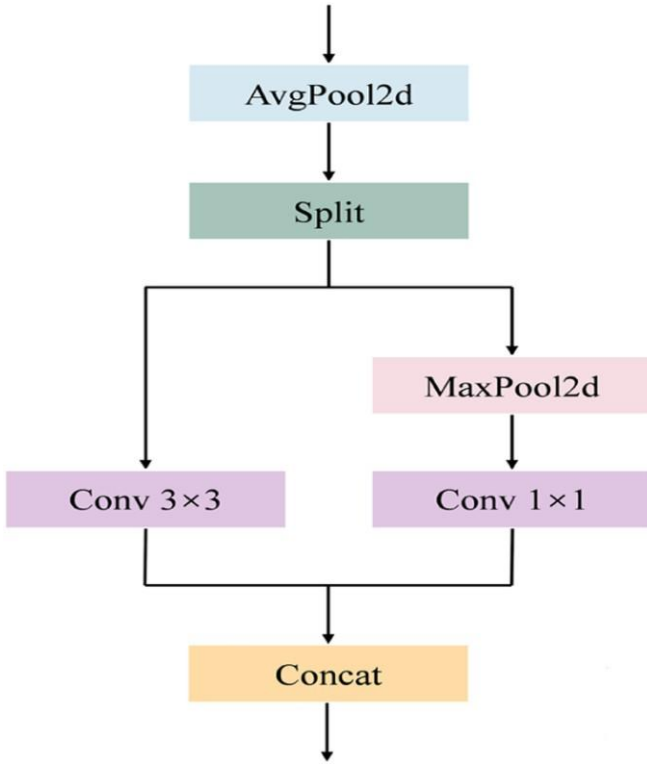


Fig: Adown Module Downsampling [1] number of parameters in the model is up to **2.14 million**, and the computational cost is reduced to **2.14 GFLOPS**, demonstrating improved efficiency compared to the baseline YOLO architecture. Experiments are conducted by varying the base channel width and GhostConv ratio, both of which directly affect the model's parameter count and GFLOPs. Table 4. Model Configuration and Computational Complexity

Model Conf.	Base Chan.	GhostConv Ratio	Param.(M)	GFLOPs
WTAD-YOLO(Baseline)	32	-	23.87	19.15
Proposed Model	32	2	15.60	13.25
Proposed Model	32	4	11.23	9.88
Proposed Model	16	2	2.14	2.14
Proposed Model	16	4	1.83	1.91
Final Proposed Model	16	2	2.14	2.14

Reducing the base channel width from 32 to 16 significantly decreased both parameters and GFLOPs, resulting in a more compact architecture. Increasing the GhostConv ratio further reduced computation, but at the risk of feature loss in deeper layers.

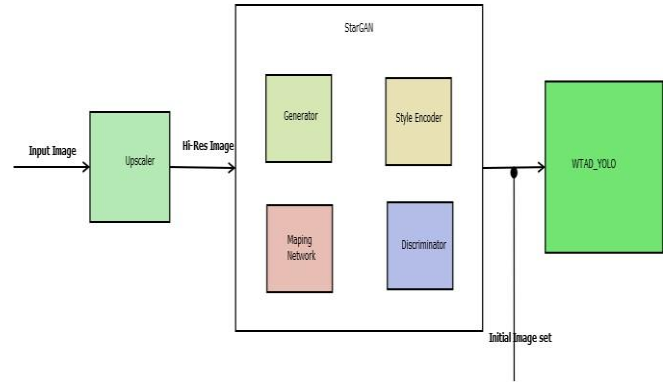


Fig: Proposed Model Architecture

This end-to-end pipeline unifies synthesis and detection into a single framework.

IV. RESULT

The proposed model successfully integrates the

GhostConv module and achieves lightweight architecture while maintaining competitive performance. The total

Access to registered agricultural datasets is restricted, the model could not be properly evaluated for detection accuracy. The configuration analysis demonstrates that the base channels= 16 and GhostConv ratio =2 setup achieved the lowest computational cost (2.14 GFLOPs) and minimal parameter count (2.14 M), confirming its suitability for lightweight and real-time agricultural disease detection applications once annotated data become available. Experiments are conducted to compare the proposed model with the baseline YOLO11 and original WTADYOLO configuration.

Table 5. Model Comparison of Computational Complexity

Model	Param.(M)	GFLOPs	Observation
YOLO11	23.87	19.15	Standard Architecture
WTAD-YOLO	2.32	6.30	Higher Accuracy with Reduced Complexity
Proposed WTADYOLO(GhostConv ratio=2,base=16)	2.14	2.14	Most efficient

The base channels from 32 to 16 and applying GhostConv (ratio=2) significantly decreased both the number of parameter and computational cost.

V. DISCUSSION

A hybrid framework combining StarGAN v2 and WTAD-YOLO is developed to address the limitations of existing agricultural disease diagnosis systems, particularly data scarcity and imbalance. By synthesizing diverse diseased leaf images from limited samples, StarGAN v2 enhances dataset diversity and enables the model to learn lesion variations and rare disease patterns, leading to improve robustness.

Extreme class imbalance in the tomato-leaf dataset illustrates where late-blight samples dominate the dataset, while healthy leaves and some diseases appear only a handful of times. By generating additional synthetic images for categories, StarGAN v2 helps smooth the distribution and provides the detector with sufficient examples to learn discriminative features. Ghost and wavelet modules can introduce slight performance degradation if annotated images are not used. Hyper-parameter tuning is essential to balance efficiency and accuracy.

The StarGAN v2 enriches data diversity by generating multiple manifestations for each disease under varying lighting and environmental conditions. Meanwhile, Ghost Conv module in WTAD-YOLO reduce parameter while keeping high accuracy. Together, Ghost Conv module and wavelet convolution module enhance generalization, minimize overfitting and balance efficiency with precision. Although the model is trained using limited synthetic and publicly available data due to restricted access, the experiment setup confirms that the model retain high detection accuracy with reduced computational complexity.

Overall, the proposed StarGAN v2-WTAD-YOLO framework demonstrates strong potential for scalable and reliable crop disease diagnosis with cross-application benefits across diverse crops and environmental conditions.

VI. CONCLUSION

This paper examines how generative models and efficient detectors can be combined to improve plant disease diagnosis. We reviewed the principles of GANs and their role in data augmentation, highlighted the diverse image synthesis capabilities of StarGAN v2 and summarized the YOLO detection framework and its lightweight variants. We describe a pipeline for generating synthetic diseased tomato leaves and training a WTAD-YOLO detector. This study shows that GAN-based augmentation can improve classification robustness, suggesting similar gains for object detection. By unifying image synthesis with efficient detection, precision agriculture can achieve more robust and scalable disease diagnosis across diverse environments.

A. LIMITATIONS

While synthetic images expand training set, there remains a risk that the distribution of synthetic data does not perfectly match real field conditions. Diseases may interact with environmental factors like humidity, lighting and occlusion in ways that GANs trained on clean images cannot reproduce.

After training, the number of parameters unexpectedly increase instead of decreasing as initially intended, indicating that further optimization of the GhostConv integration is required.

Additionally, access to real-world agricultural datasets is restricted due to dataset registration and usage policies, limiting the model validation on large-scale field data. The main limitation is that, we are not used annotated dataset as original dataset are annotated. But we cannot access it. For that, we can not find mAP and F1-score.

B. FUTURE WORK

Future work will focus on improving the efficiency of the GhostConv integration to ensure that parameter reduction is achieved without compromising accuracy. Advanced optimization techniques such as layer pruning, quantization or hybrid lightweight modules will be explored to minimize

computational cost. To overcome the limitations of synthetic data, domain adaptation and transfer learning approaches will be applied to better align GAN-generated samples with real field conditions. Moreover, efforts will be made to collaborate with agricultural research institutions to gain access to registered real-world datasets, enabling large-scale validation and enhancing the models generalizability under diverse environmental conditions.

VII. REFERENCES

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